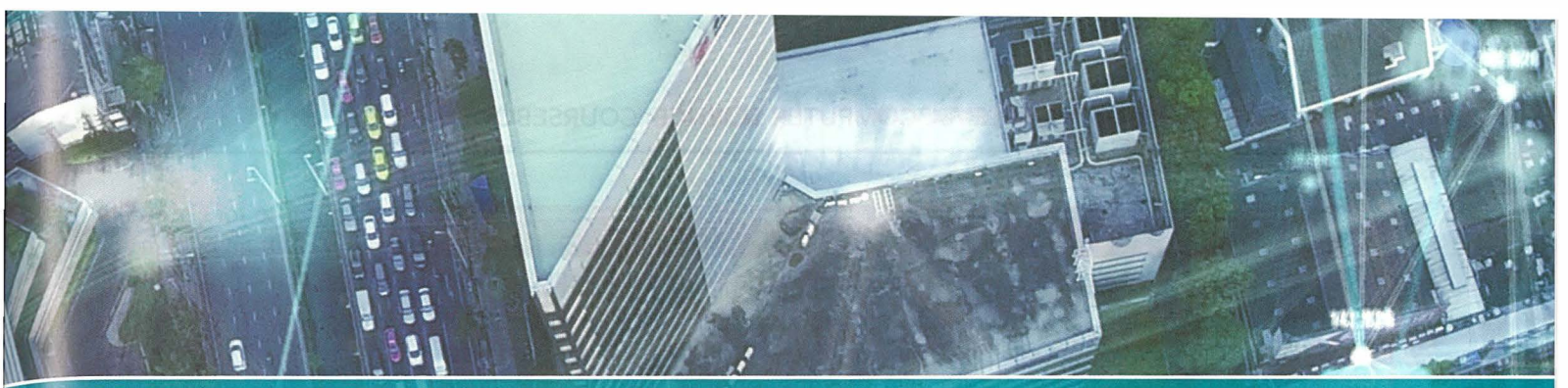


Chapter 6
Automated and emerging technologies

IN THIS CHAPTER YOU WILL:

- learn about the use of automated systems in context
- explore how automated systems use sensors, microprocessors and actuators
- explore the advantages and disadvantages of automated systems in context
- learn about the use of robotics
- learn about the characteristics of a robot
- explore the use of robots in context
- learn about what is meant by artificial intelligence
- explore the characteristics of artificial intelligence systems
- describe the use of machine learning in artificial intelligence
- explore the features of, and use of, expert systems.

GETTING STARTED

Design a robot. Consider a scenario where a robot would be useful in your daily life. Identify the tasks you would like the robot to perform. Describe how the robot will work, for example, the inputs that would be used (and the hardware, e.g. sensors that might do this), how the robot will decide what to do, and what its output will be.

WHAT IS INTELLIGENCE?

There are many ideas on what it is to be intelligent, with one key area being the ability to learn. How did you develop intelligence? From when you were very young you started to learn. Each time you experienced something new, you learnt from it, for example, if you cried then someone might pick you up. So, next time you wanted to be picked up, you cried.

The desire to create intelligent computers has been around since the very first machines. Even looking back to ancient history there are stories about bringing objects to life.

Achieving artificial intelligence has made the most steps over the last 20 to 30 years. There are now systems that simulate some human activities, and systems that can learn both with and without a person telling them what to learn. These programs all still run on rules, that are programmed into them to follow, so are they intelligent? Humans learn by developing rules and storing these in memory, so are we intelligent?

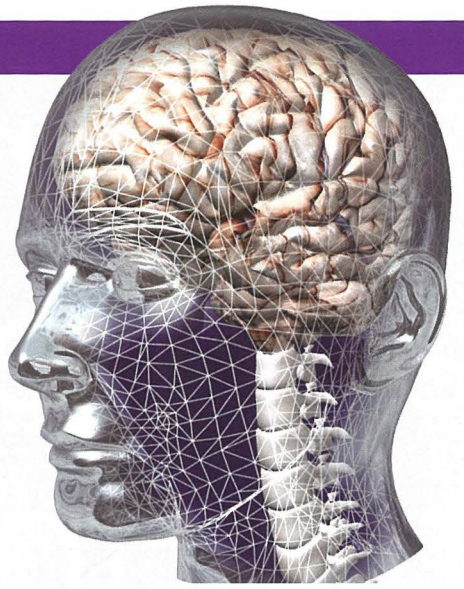


Figure 6.1: A brain

Discussion questions

1 What do you think is meant by intelligence?

2 How would you test whether a computer is intelligent, or not? What would you test it against?

3 Do you think that computers that learn and act independently should be created?

6.1 Automated systems

An **automated** system performs actions without interaction with humans. They are used in small systems such as the heating or air-conditioning in buildings, all the way up to flying aeroplanes and building items such as cars.

KEY WORD

automated: a device that is operated without human interaction.

Features of automated systems

An automated system usually has the following features:

- Sensors – to measure the environment.
- Microprocessor – to process the data from the sensors and determine if an action needs to be taken.
- Actuators – to create a movement.

There are a lot of different **sensors** that you need to know about – you can find more information on these where they were covered in Chapter 3, Section 6. You will need to identify which sensor is most appropriate in a given context.

For example: a system needs to monitor the pollution in a river.

Consider what could be measured. The temperature of the water using a temperature sensor. The amount of light reaching the bottom using a light sensor. The pH of the water using a pH sensor.

Sensors just do one job, they read the environment and send the data onwards. They do not process the data. That is done by the **microprocessor**. There is a problem though, the sensor reads **analogue** data (real-world data such as 23.5 °C), but the microprocessor only understands binary (01101010). The sensor first sends the data to an analogue-to-digital converter, to turn the reading into binary (digital). Then, the **digital** data is sent to the microprocessor.

Before the system is started, data will be stored in the microprocessor. This could be a single value, for example, the value from an infra-red sensor to state that there is movement. There could be several values, for example, the minimum temperature and maximum temperature for a greenhouse. Or there could be a whole range of data.

The microprocessor takes the data from the sensor and compares it to its value, or values. The result of the comparison determines the next step. For example, in a burglar alarm, if the sensor is outside of the stored range then an alarm will need turning on. In a house, if the temperature is below the minimum the heating turns on, and if it's above the maximum it turns the heating off.

If the microprocessor decides an action is needed (e.g. turn the heating on), it sends a signal to the **actuator**, which then performs the action. The whole process is then repeated until the system is turned off.

KEY WORDS

sensor: a type of input device that is used to capture data from its immediate environment.

microprocessor: an integrated circuit that is able to perform many of the functions of a CPU.

analogue: continuous data that can be any value within a range.

digital: discrete data that is stored as 1s and 0s.

actuator: a mechanical part that causes another device or part to move.

Example

A set of doors in a building automatically open when a person is detected.

1 What type of sensor could be used?

An infra-red to detect movement, or a pressure sensor to detect if someone is standing near it.

2 What happens to the analogue data?

The analogue data is converted into digital.

3 Where does the data go?

The data is sent to the microprocessor.

4 What happens in the microprocessor?

It compares the received data with the stored data. If it is out of range, it sends a signal to an actuator to open the doors. If it is not out of range, it does nothing.

This process continues until the system is turned off.

This can also be shown in a diagram in Figure 6.2:

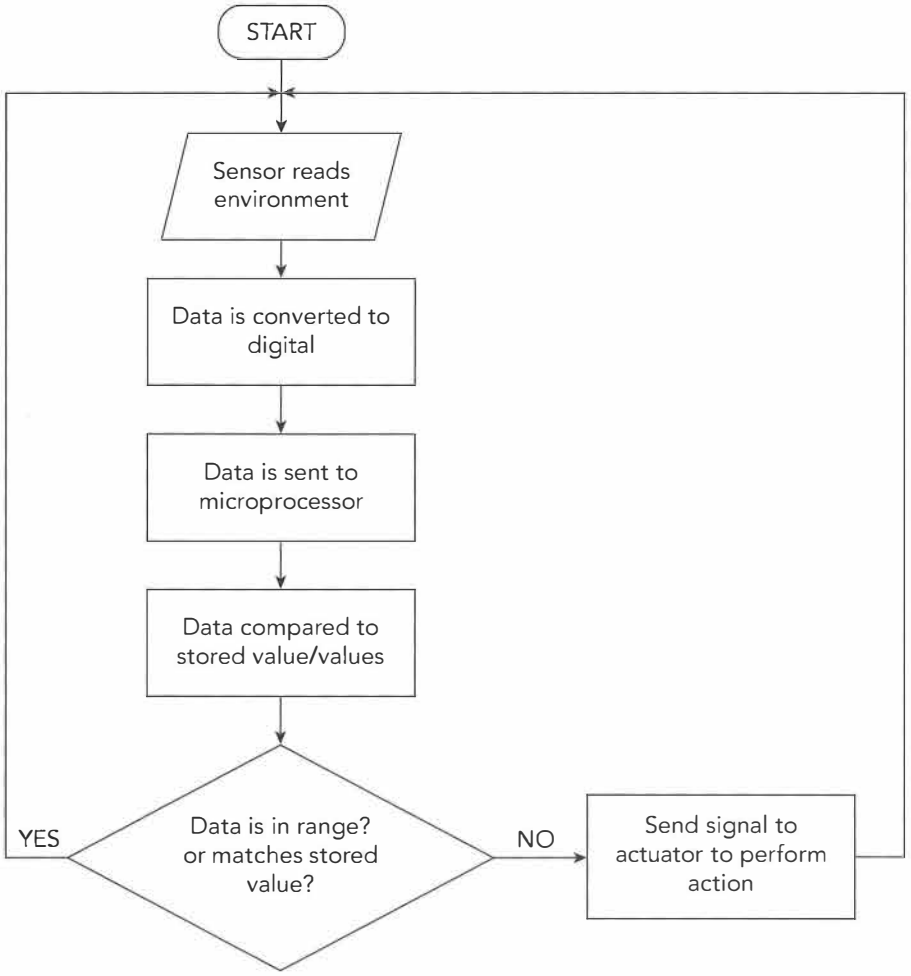


Figure 6.2: The process in which doors open automatically when a person is detected

COMPUTER SCIENCE IN CONTEXT

Control systems are at work all around you, probably without you noticing. Think about a car. It could have light sensors, so when it is getting dark the headlights will be automatically turned on. It could have rain sensors, so when it starts raining the windscreen wipers start moving. These each have a sensor that is constantly reading the values of the environment, e.g. the light level. The data will be sent to a microprocessor that compares the value with its stored value, and if necessary sends the signal to turn on the lights, or the wipers. This means the car drivers do not have to worry about turning these on when needed. It also improves safety, particularly with the headlights so that the car is always visible.

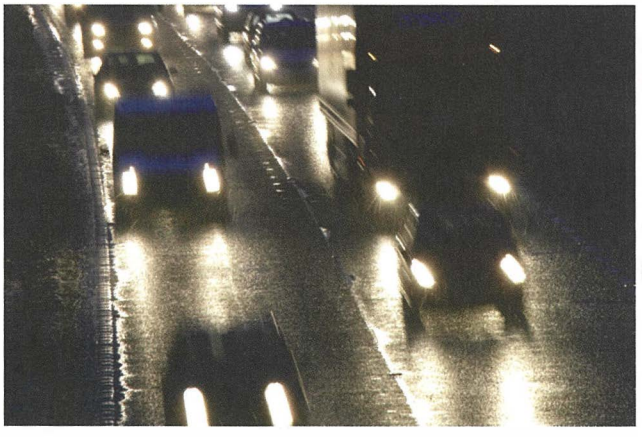


Figure 6.3: Car headlights could operate via sensors

SKILLS FOCUS 6.1

CONTROL AND MONITORING SYSTEMS

When considering a control or monitoring system you need to determine the inputs, processes and outputs as you would in a computer program:

- The inputs from the sensors.
- The processes.
- The outputs from the system.



Figure 6.4: A control or monitoring system can be thought of as a 'story' of input, process and output

Using these you then need to create the 'story' (see Figure 6.4) about how the sensors gather the appropriate data, what this is compared to, and how this influences the outputs.

For example:

An automatic door will open when a person is detected. The door stays open for 5 seconds if no more people are detected.

Take each point in turn:

- Inputs:

  A pressure sensor can be used to detect if a person is standing on a specific point.
  A motion or infra-red sensor can be used to detect if a person is in a specific place.

- Processes:

  Analogue data from the sensor will be converted to digital.
  The microprocessor will store the value where a person is detected on the pressure sensor, and/or the result identifying that a person (movement) has been detected.

- Output:

  A signal will go to an actuator to open the door.
  This will stay for 5 seconds, and is reset each time another person is detected.

Questions

1 Describe how the sensor will monitor its environment.
2 Describe how the processor will work out if the door needs to be opened.
3 When will the door close?

Questions

1 What type of data does a microprocessor process?
2 What tasks does a sensor perform?
3 An automated system makes cakes by positioning a cake tin below a funnel where the cake mix comes through to fill the tin.
   a Identify a suitable sensor for this system.
   b The sensor has read the data. Identify the function that happens next.
   c Where is the data sent?
   d Describe the actions the microprocessor performs.
   e How many times does the system run?

Automation in context

KEY WORD

automation: the use of automated equipment in a process, e.g. a factory.

The **automation** of systems has both benefits and drawbacks, and these are different depending on the system. You need to be able to think about the context and the benefits and drawbacks in that scenario.

When evaluating the use of automation you can consider:

- initial cost
- running cost
- safety
- replacing people's jobs
- continuous work all day every day
- precision.

Example 1, industry

Car manufacturing.

- Initial cost: this will be high because the robots need purchasing. The software will also need developing and could be expensive.
- Running cost: this could be high because the systems need maintenance, but it could perform actions more efficiently than humans which could save money. However, there may be fewer employees, therefore saving money long-term.
- Safety: it should be safer because they can monitor their environment and stop if there is any danger, whereas people can be distracted, and more accidents could happen.



Figure 6.5: A car manufacturing factory

- Replacing people's jobs: jobs that the system now performs will be lost, but more jobs will be made to maintain the system.
- Continuous work all day every day: this would mean that more cars can be built.
- Precision: human errors will not be made so there are less faulty cars.

Example 2, science

Nuclear power plant.

- Initial cost: will be high because of the purchase of the devices and development of the system.
- Running cost: may be lower because fewer people are needed, however maintenance might be expensive.
- Safety: less risk of human error therefore increasing the safety, also people will not need to work in dangerous places.
- Replacing people's jobs: some jobs will be replaced for the functions the system does, but new jobs will be made to maintain the system.
- Continuous work all day every day: it is important that the power plant is monitored all day every day without getting tired and therefore less errors.
- Precision: this is important in a high-risk scenario and an automated system will measure and perform actions with more precision, therefore fewer errors.

Example 3, agriculture

Growing conditions within a greenhouse.

- Initial cost: this may be high for the installation and programming.
- Running cost: there will be a cost for electricity and maintenance but it is a small system so may not be high.
- Safety: this may not be relevant because it is not a dangerous process.
- Replacing people's jobs: it might replace jobs of people who monitored the conditions, but this is unlikely to be a large number.
- Continuous work all day every day: this will allow the system to adapt as soon as there is a problem, instead of waiting for a person to check and make changes.
- Precision: this may not be as important, but it will allow readings to be more accurate and instant changes to be made.



Figure 6.6: A greenhouse

Questions

4 An auto-pilot system will fly an aeroplane without input from a person.
Identify **two** benefits and **two** drawbacks of using the auto-pilot.

5 A weather station records and stores data about the current weather conditions.
Describe the advantages and disadvantages of using the automated system.

6.2 Robotics

What are robots?

A **robot** is a mechanical device that performs an action, usually an action that a human would perform. **Robotics** is an area that covers the design, construction and operation of robots.

A robot can be controlled by a person, or they might be automated, or a mixture of the two.

One example is a drone. This is a robot, it can be controlled by a person, and it can be programmed to be perform movements depending on its surroundings.

Another example is the machines that build components in industry, for example, manufacturing devices. The robots that put the components together can be automated so that they can build the components without human interaction.

Robots have a range of features:

- A mechanical structure or framework. This is the body of the robot.
- Electrical components such as:
  - Sensors to record its environment, for example, the position of the component it is building.
  - Microprocessors, to take the reading from the sensor and decide the action to perform.
  - actuators, to make the robot move.

Robots are also **programmable**; a program can be written for the robot to follow.

KEY WORDS

robot: a machine that replicates human actions or movement.

robotics: an area of computer science that looks at the creation and use of robots.

programmable: a computer that will run the commands stored in it.

6.3 Robots in context

Robots in medicine

Developments include the use of robot nurses that can move around a hospital to visit and interact with patients. This can save the doctors and nurses time so that they can spend more time doing other tasks.

Robots are also used to perform procedures such as operations. For example, if a surgeon is not present in the hospital, even on the other side of the world, they can control the robot performing the procedure. This means that a patient will not have to wait for their procedure, and specialists are not required in every hospital because they can share resources.

The use of robots in medicine also has drawbacks. If there are any errors in the programming, or someone gains access to the programs and changes them, then the result could cause harm. If a surgeon is controlling a robot from a different location and the network connection is lost, then the procedure will stop. This means there needs to be backup systems ready to work if there are any problems.

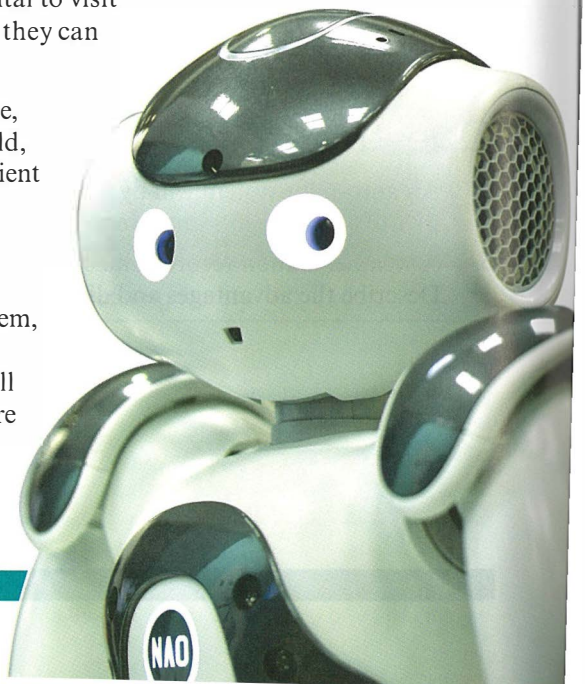




Figure 6.7: Robots preparing for surgery

Robots in agriculture

Developments include automated tractors and other machines, they can make use of satellites and satellite navigation to guide their movements and then perform the required actions. This means that a farmer can be performing other actions, whilst the machines perform the repetitive tasks that would take up their time otherwise.

Robots can also be used to plant seeds, remove weeds, check and adjust the environment where the plants are growing. All of these can be carried out 24 hours a day, 7 days a week, whereas a person may only be working 8 hours a day.

Robots in agriculture are often used for the repetitive tasks that people would otherwise spend a lot of time performing. The upfront cost might be high, and there will be maintenance required. However, it might also replace a worker who would otherwise be performing these actions.



Figure 6.8: A lawnmower working without a driver

Transport

A self-driving car is a robot. It reads its environment and moves accordingly. This is still being developed, but eventually people may not need to drive cars, this could improve safety because the robot can react faster than a person. It is an expensive up-front cost, and there is the threat of what happens if someone accesses the program of a car and changes it.

Robots are commonly used in factories to transport items. For example, in a factory, robots can move boxes of product to the lorries. These can be completely automated, keeping people away from the movement of products that could be dangerous. They do replace people's jobs, but also provide new jobs to run and maintain the system.



Figure 6.9: A self-driving car

Industry

Industry is the manufacture of goods from raw materials. This could be from the building of a pencil, to the building of an aeroplane and everything in-between.

Some factories are completely automated with robots. A car production line can involve many different robots, each performing a specific role. The car being built moves from one robot to the next without the need for human interaction. This means that the factory can run all day, every day. The robots do not need to rest and can be more precise because they are not prone to human error. However, their roles were previously carried out by people who may now be without a job. Instead a smaller number of people are required to monitor and maintain the robots, with a different skill set than those who have been replaced. The initial cost is also very large as the equipment needs to be purchased, and the program produced.

Robots don't always manage all parts of a process. Some robots can:

- Manage the packing of items, for example, putting packets of biscuits into boxes of 24 packets.
- Test products, for example, checking the temperature of products when they come out of an oven.
- Perform tasks that are minute in size and require exact precision, for example, the building of circuits.

Entertainment

There are many robots that can be bought as toys, such as robot dogs for people to interact with. There are also those that are educational, for example, where a child can build a robot and then program it to perform specific functions. These allow people to engage with technology, and in some cases learn about how robots work and be introduced to programming.

COMPUTER SCIENCE IN CONTEXT

Drones are another example of where robots are used in entertainment. People can use their drones to record videos of themselves performing tasks. These allow people to get a birds-eye view of an area, but a loss of contact between the drone and the remote could cause the drone to crash and potentially cause injury to people nearby.



Figure 6.10: A drone

Domestic

Domestic robots are those that perform tasks around a home. The most common one is the robot vacuum cleaner. It will make its way around a room, turning when it detects obstacles and changing direction. This can save people time as they will not need to perform the task themselves, however they still need monitoring and cannot go up and down steps. A robot lawnmower works in the same way, monitoring the area it is mowing to make sure it does not miss any spaces.

COMPUTER SCIENCE IN CONTEXT

A security robot can be used to protect a house. It could record what is happening as videos, and detect motion or sound. These can be set to patrol an area and alert someone if an abnormality (something unusual) is detected.

ACTIVITY 6.1

Find more examples of robots that are used in each of the sections; medicine, agriculture, transport, industry, entertainment and domestic. List the benefits and drawbacks of each of the robots. Share your findings with the class and see how many different robots you have all found.

REFLECTION

After completing Activity 6.1, it can be helpful to think about how you approached it and ways you could improve your research skills for future activities.

1 Discuss the sources you used to find examples of robots. How did you make sure these sources were reliable, up-to-date and accurate? What would you do differently if you had to find further examples?

2 Discuss whether finding your own examples might help you answer exam questions. Is it helpful to find your own contexts instead of relying on those given in class?

Questions

6 Define the term robot.

7 Identify **two** ways robots can be used in industry.

8 Give **one** benefit and **one** drawback of the use of domestic robots.

9 Discuss the use of driverless cars.

6.4 Artificial intelligence

Can computers think? This is a question that has been asked and sought after for many years. Alan Turing developed a test in 1950, called the Turing Test, to assess whether a computer can display intelligence. The test involves a person holding a conversation with a computer program, and another person. If they cannot work out which is the computer and which is the person they are talking to, then it is has passed the Turing Test. Some people do not think that this would demonstrate intelligence, the computer is still following a set of rules that are pre-written.

What is artificial intelligence?

Artificial intelligence (AI) is the development of programs to simulate human behaviour. There is a broad range of areas within AI:

- Image recognition; to identify objects or people in an image.
- Speech recognition; to identify words spoken and store them.
- Natural language; to receive a command or instruction that is not in a set format and perform the required response.
- Computer games; to move elements or characters independently based upon the environment.
- Diagnosis systems; for example, medical diagnosis.

You have probably used natural language systems, for example, on a smartphone or a web browser. You can enter your request in dozens of different ways. Let's say you want to find out what time the sun sets today. You could enter "sunset today time", or "what

KEY WORD

artificial intelligence (AI): a part of computer science that looks at creating machines that can think and perform tasks a person would usually perform.

time does the sun set today?" or "today when does the sun set?" or many other ways. The software needs to understand what you have asked to decide how it will answer.

Some simple questions could be hard-coded. That means every possible way of asking a question is written in the program, so it just has to select the correct one. The problem is, how many ways are there? Is it possible to hard-code every possible question or statement that anyone in the world could ever give? This would be impossible. So instead, the artificial intelligence program has to be able to adapt to whatever is entered. This could be done by comparing it with similar phrases, or picking out key words and grammar, or it could use neural networks which attempt to simulate the human brain to process the words entered.

ACTIVITY 6.2

Identify any programs you have used that might make use of artificial intelligence. This could be computer games where you play against the computer, or natural language interfaces. Discuss how you think these may have been programmed and how the problems can be simplified into smaller parts to make the task easier to tackle.

Components of AI

Artificial intelligence programs can be simplified to the four key features:

1. **Collection of data:** A program will need data input, this might come from a user (e.g. speech recognition) or from its sensors (e.g. a robot that uses infra-red sensors to work out its position and where it can and can't move).
2. **A set of programmed rules**: These are stored for the program to use to make decisions. For example, a calculation to determine which decision to make. A rule that if an object is in front of the sensor, turn to the right.
3. **The ability to reason:** Reasoning is an area of logic, where you have rules, and from these you can develop facts. For example, all dogs eat meat. Fred is a dog. Therefore, Fred must eat meat.
4. **The ability to learn and adapt:** Not all AI systems will do this, this is a specialist area known as machine learning. Systems can be programmed to learn, and change their own rules and logic. This often involves training. If you want a program to differentiate between an image of a dog and a horse, you can train the system by showing it images of dogs and horses and telling it what the images are showing. The program will then learn the characteristics of a horse and a dog, so when it is tested with a new image it will hopefully identify the animal.

Machine learning

Some AI systems do not learn, for example, in a computer game the characters can move and interact with your character and adapt to what you do, but they are not learning what you do, they are following their pre-programmed rules.

Machine learning can be supervised, or unsupervised.

Supervised means that user is telling the program what its data means, for example, in the image recognition example the user says this image shows a horse.

KEY WORD

machine learning: a computer program that can adapt its stored rules or processes.

Unsupervised means that data is input, and then the program learns from the data. An example of this is clustering. The data is plotted on a graph, and then the program identifies which items of data are close to each other. In image recognition, the shapes in the images could be input and then the features plotted, the program can then group the images according to these features. This is done without any human interaction needed.

COMPUTER SCIENCE IN CONTEXT

Components in a computer game can make use of machine learning. For example, an AI chess player. The program will alter the moves that it makes based on the games it plays. This simulates how a human learns to play the game, they learn which moves work and which don't. This could be supervised learning if they are told which moves are the best to make. It could be unsupervised learning where it can play other people or computer programs, and change its own data and rules without being told what to do.

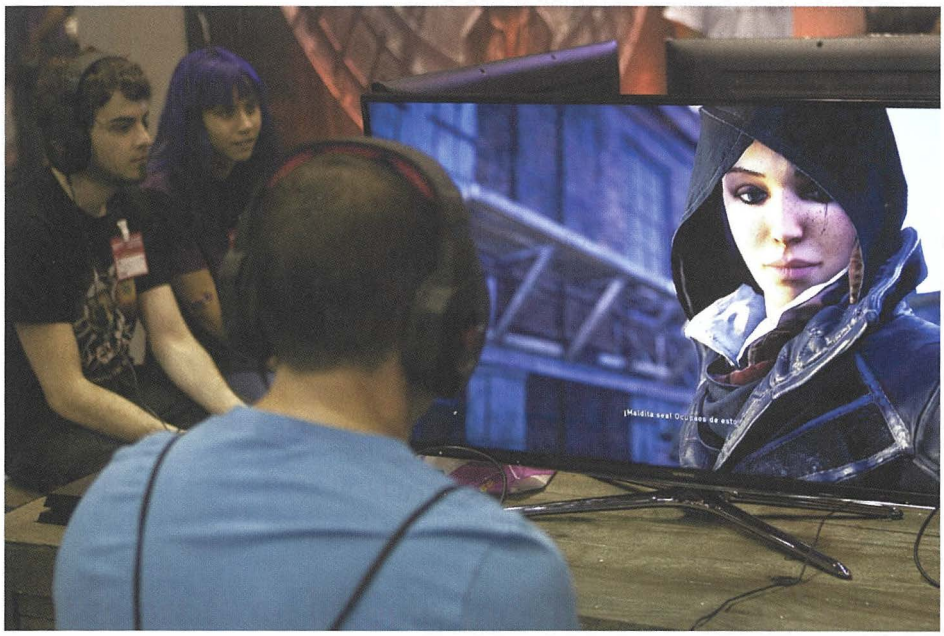


Figure 6.11: A person playing a computer game

ACTIVITY 6.3

Find and play a computer game where you are competing against the computer. Discuss how you think the computer's character was controlled. Did it just repeat the same movements over and over, or did it change what it did based on your move? Discuss how you think it was created and programmed.

REFLECTION

After completing Activity 6.3, it can be helpful to think about how you approached it and ways you could improve your research skills for future activities.

1 How did you approach this task? Did you play the computer game without thought to the character? Did you try lots of different moves to see how the computer reacted?

2 What programming knowledge did you use to discuss how the computer's character was programmed?

Expert systems

An **expert system** is a special type of system that attempts to emulate the expertise of a human, for example, an engineer, or doctor. The system asks the user questions to determine the solution or answer. Depending on each answer, the system will ask a different question.

For example, a program to help you fix a printer might first ask:

'Is there a green light on the printer?'

If the user says No, then they may be told, 'Press the button to turn the printer on.'

If the user says Yes, then they may be asked, 'Is there a red light underneath the paper picture?'

An expert system has four key features:

1 A **knowledge base**. This is a list of facts, for example, there is a green light below the power symbol.
2 A **rule base** that links the facts. For example, If the green light is on then the printer has power.
3 **Inference engine**. This is the part of the program that decides which question to ask next, or which answers it gives.
4 The user interface. This is the part that outputs questions and statements to the user, and allows the user to enter data.

KEY WORDS

expert system: a system that attempts to replicate the knowledge of an expert.

knowledge base: part of an expert system that stores the facts.

rule base: part of an expert system that stores the rules based upon the knowledge.

inference engine: part of an expert system that makes the decisions.

COMPUTER SCIENCE IN CONTEXT

Organisations often use expert systems to allow people to find help without having to communicate with people. These are usually repetitive problems that people may encounter. If you have a home internet connection and it is not working, then troubleshooting can tell you what actions to perform to fix it.

ACTIVITY 6.4

Create an expert system to determine the difference between a cat and a dog. Ask the user questions depending on their previous answers. This can be written as a computer program using a programming language of your choice, or as a flowchart showing which questions to move to depending on each answer.

Peer Assessment

Test each other's expert systems. How useable was the system? Were there any questions that you couldn't answer? Did it give you the correct answer?

Questions

10 Define the term artificial intelligence.

11 Describe what is meant by machine learning.

12 State the purpose of an expert system.

13 Give **two** examples of where expert systems can be used.

14 An expert system has an inference engine and interface. Describe the **two** other component parts of an expert system.

SUMMARY

An automated system performs actions without human intervention.
An automated system has sensors, a microprocessor and actuators.
Robotics is the creation and management of machines that perform actions that humans perform.
A robot contains a mechanical structure, electrical components including parts of an automated system, and it can be programmed to perform specific actions.
Robots can be found in a range of areas, for example, medicine and transport, and have advantages and disadvantages in each context.
Artificial intelligence is an area of computer science that aims to develop systems that mimic human intelligence.
Machine learning is one area of artificial intelligence where a program can amend its own data and algorithms.
An expert system attempts to simulate a human expert in a specific area.
An expert system is made up of a knowledge base, rule base, inference engine and interface.

EXAM-STYLE QUESTIONS

1 a **Give** *one* example of where an expert system can be used. [1]

b Complete the sentences about expert systems by filling in the missing words.

An expert system is made up of four parts. The ..... base is the set of ..... The ..... base is a set of logical rules. The .... applies the two bases to decide what questions to ask next and to work out the response. The ..... is the elements that allow the user to ..... data and to output data to the user. [7]

[Total: 8]

2 **Identify** *three* characteristics of a robot. [3]

3 A car has a proximity detection system. If an object is within a set distance from the car, an alarm is sounded in the car.

Describe how the automated system detects objects and alerts the driver. [6]

4 An artificial intelligence program uses machine learning.

Describe machine learning using an example. [2]

5 A factory that builds furniture is thinking about buying robots to replace the workers.

Discuss the advantages and disadvantages of the factory replacing the workers with robots. [6]

6 Identify *two* ways that robotics can be used in transport. [2]

7 Describe how machine learning can be used to develop speech recognition systems. [4]

8 A farmer is thinking about implementing software into his machines to turn them into automated systems, for example, a tractor that navigates its own way around the field to plough the field.

a Identify *one* other automated system the farmer could make use of. [1]

b Give *two* benefits to the farmer of using automated systems. [2]

c Give *two* drawbacks of implementing automated systems in the farm. [2]

[Total: 5]

9 A film company want to video animals in the wild acting naturally. They decide to create life-like robots that use artificial intelligence to react naturally with their environment. The robots have cameras inside that record the interactions.

a Describe how artificial intelligence can be used to allow the robots to react to their environment. [6]

b Discuss the use of robots to record the animals. [6]

[Total: 12]

COMMAND WORDS

give: produce an answer from a given source or recall / memory.

identify: name / select / recognise.

describe: state the points of a topic / give characteristics and main features.

SELF-EVALUATION CHECKLIST

After studying this chapter, think about how confident you are with the different topics.
This will help you to see any gaps in your knowledge and help you to learn more effectively.

You might find it helpful to rate how confident you are for each of these statements when you are revising.
You should revisit any topics that you rated 'Needs more work' or 'Getting there'.

I can...	See topic	Needs more work	Getting there	Confident to move on
describe what an automated system is.	6.1			
describe the hardware used within an automated system.	6.1			
describe the role of sensors, microprocessors and actuators used in automated systems.	6.1			
describe the advantages and disadvantages of an automated system in a given scenario.	6.1			
define the term robotics.	6.2			
give examples of the use of robotics.	6.2			
describe the characteristics of a robot.	6.2			
describe the advantages and disadvantages of robotics in a given scenario.	6.3			
define the term artificial intelligence.	6.4			
describe the main characteristics of an AI system.	6.4			
describe machine learning.	6.4			
describe the key features of an expert system.	6.4			
describe the use of an expert system for a given scenario.	6.4			